Figure 2: Several attack vectors persist despite the advances in cybersecurity (Credit: Palo Alto Networks)

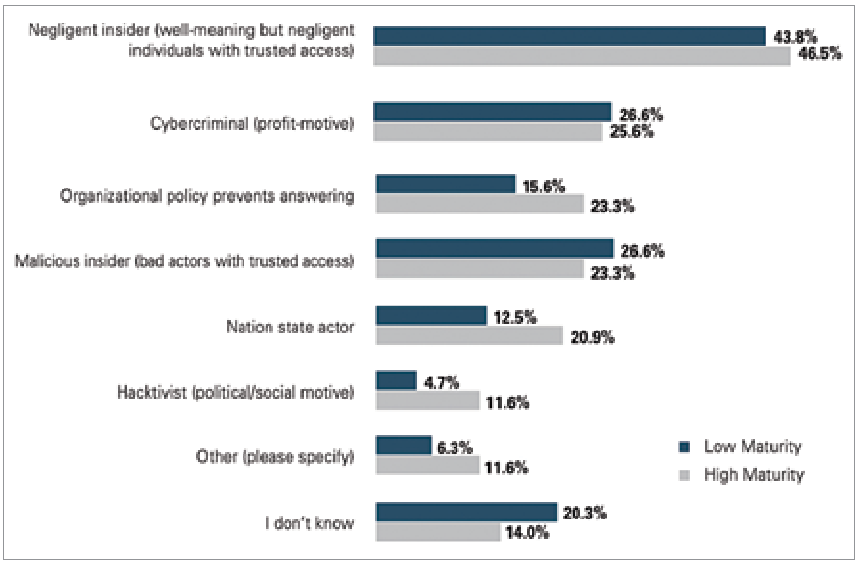

Figure 3: Threat actors of IIOT (Credit: Palo Alto Networks)

where malware is crafted. The attackers then target specific individuals, often through social engineering tactics. Delivery and exploitation follow, with malware being introduced and exploited. Installation of malicious software within the OT network occurs before the attackers execute the attack, exfiltrate data, and compromise the control system.

Overcoming security challenges

While recognising the importance of security in Industrial IoT is a crucial first step, implementing effective measures can be challenging. Several factors contribute to the complexity of securing these environments:

Legacy systems: Many OT systems are built on legacy infrastructure, making it challenging to integrate modern security solutions seamlessly.

Lack of visibility: Managing and monitoring the vast number of assets in Industrial IoT environments can be overwhelming, leading to blind spots in security.

Compatibility issues: Existing security tools may not be compatible with OT systems, necessitating tailored solutions.

Security strategy: A well-defined security strategy, starting from the top down, is often lacking, resulting in inadequate protection.

Governance and monitoring: Consistent security governance and

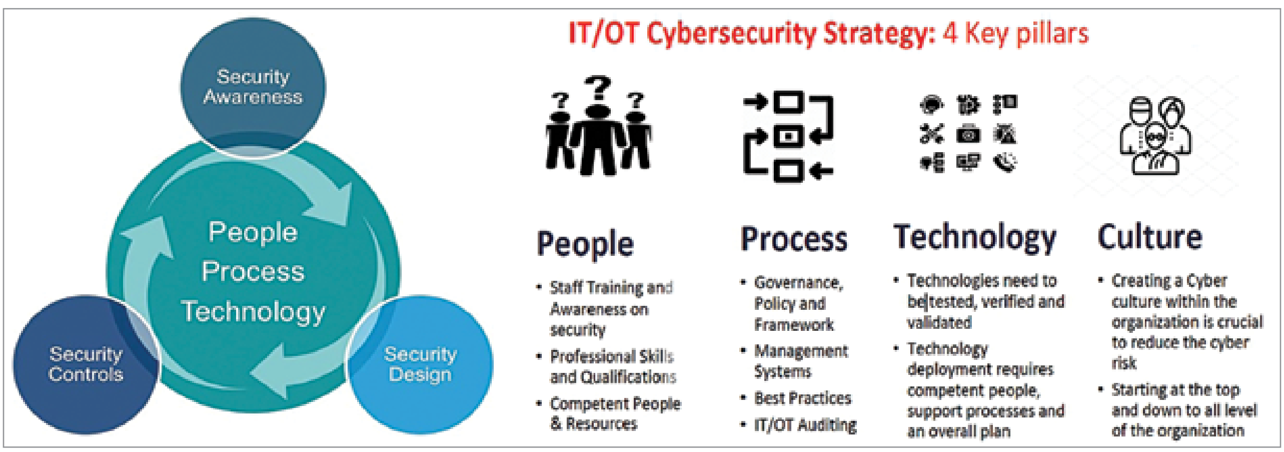

Figure 4: The four key pillars of cybersecurity strategy (Credit: Palo Alto Networks)